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An artificial intelligence model for sand and dust storm forecast driven by AI weather forecasts

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We present AI-DUST, a deep learning model for dust forecasting directly driven by AI-generated weather forecasts. Integrating a Multiple Stacked Graph Attention Network with physical constraints and a physics-based emission scheme, AI-DUST captures key atmospheric physical processes without relying on traditional numerical dust modeling chains. The model demonstrates exceptional accuracy in reproducing a traditional dust model, with correlations >0.99 (one-step) and >0.61 (80-step). In real-time forecasts of 2025 spring sand and dust storms (SDS) over East Asia, AI-DUST outperformed operational models, achieving a 27% higher Threat Score (TS) in 48-hour predictions across 14 strong events. Its 10-day forecast TS exceeds 0.22, demonstrating strong long-term capability. The model generalizes well to unseen regions like the Sahara, enabled by its architecture and standardized preprocessing. This work demonstrates the feasibility of building atmospheric environmental forecasting systems directly driven by AI-generated weather forecasts, paving the way for new, efficient AI-driven chemistry and transport models.

Sand and dust storms (SDSs) are severe meteorological events prevalent in arid and semi-arid regions, with significant impacts on climate change, atmospheric conditions, human health, and socioeconomic activities[1]. East Asia ranks among the regions that are most frequently impacted by such events[2]. Notably, strong SDS events in 2021 and 2023 have caused substantial environmental and economic impacts across the region[3,4]. Accurate SDS forecasting is therefore critical for mitigating associated hazards[5]. Numerical dust models, which explicitly represent key physical processes such as emission, transport, and deposition, have become the primary tool for dust forecasting[6]. Nevertheless, these models continue to face challenges in terms of forecast accuracy and computational efficiency[7].

With the rapid advancement of artificial intelligence (AI), its applications in weather forecasting have become increasingly widespread[8,9]. Machine learning has also been widely adopted for dust weather prediction. One approach uses machine learning to correct biases in traditional numerical model simulations for SDS at specific sites[10,11]. This hybrid approach has demonstrated significant improvements in forecast accuracy of SDSs forecasting over China[11]. However, these methods remain inherently dependent on the simulations of numerical dust forecasting models, which are computationally expensive. Alternative approaches leverage satellite observations for short-term SDS forecasting, achieving good performance within 24 h but limited skill for longer lead times[12]. For instance, utilized deep learning to predict dust storm movement directions from satellite imagery. More recently, end-to-end AI frameworks have been developed to jointly predict atmospheric aerosols and meteorological fields[13]. Gui et al. developed an AI-driven global aerosol-meteorology forecasting system, providing reliable 5-day, three-hourly forecasts of aerosol optical components and surface concentrations[14]. Bodnar et al. introduced Aurora, a large-scale foundational model for Earth systems, which outperforms operational air quality forecasts[9]. However, such models face challenges, including the sample imbalance problem in training data[15] and high training resource requirements. Therefore, while AI offers a promising pathway toward accuracy and scalable SDS forecasting, further research is needed to enhance both the efficient and extended lead-time skill of machine learning models, particularly for rare and severe dust events.

AI-based forecasting models have demonstrated significant potential in improving both the accuracy and efficiency of weather predictions, yet they still fall short of matching the physical comprehensiveness of traditional numerical weather prediction (NWP) models. Recent studies have explored downscaling or driving regional mesoscale meteorological models using AI-based global weather forecasts to achieve higher-resolution forecasts[16,17]. However, there remains a lack of SDS forecasting systems that directly utilize AI-generated weather prediction products. Meanwhile, machine learning has been successfully applied to emulate atmospheric chemical processes or replace atmospheric chemical transport models, significantly enhancing computational efficiency while maintaining strong predictive performance[18–21]. Although these machine learning-augmented models still rely on meteorological inputs from NWP models, they provide

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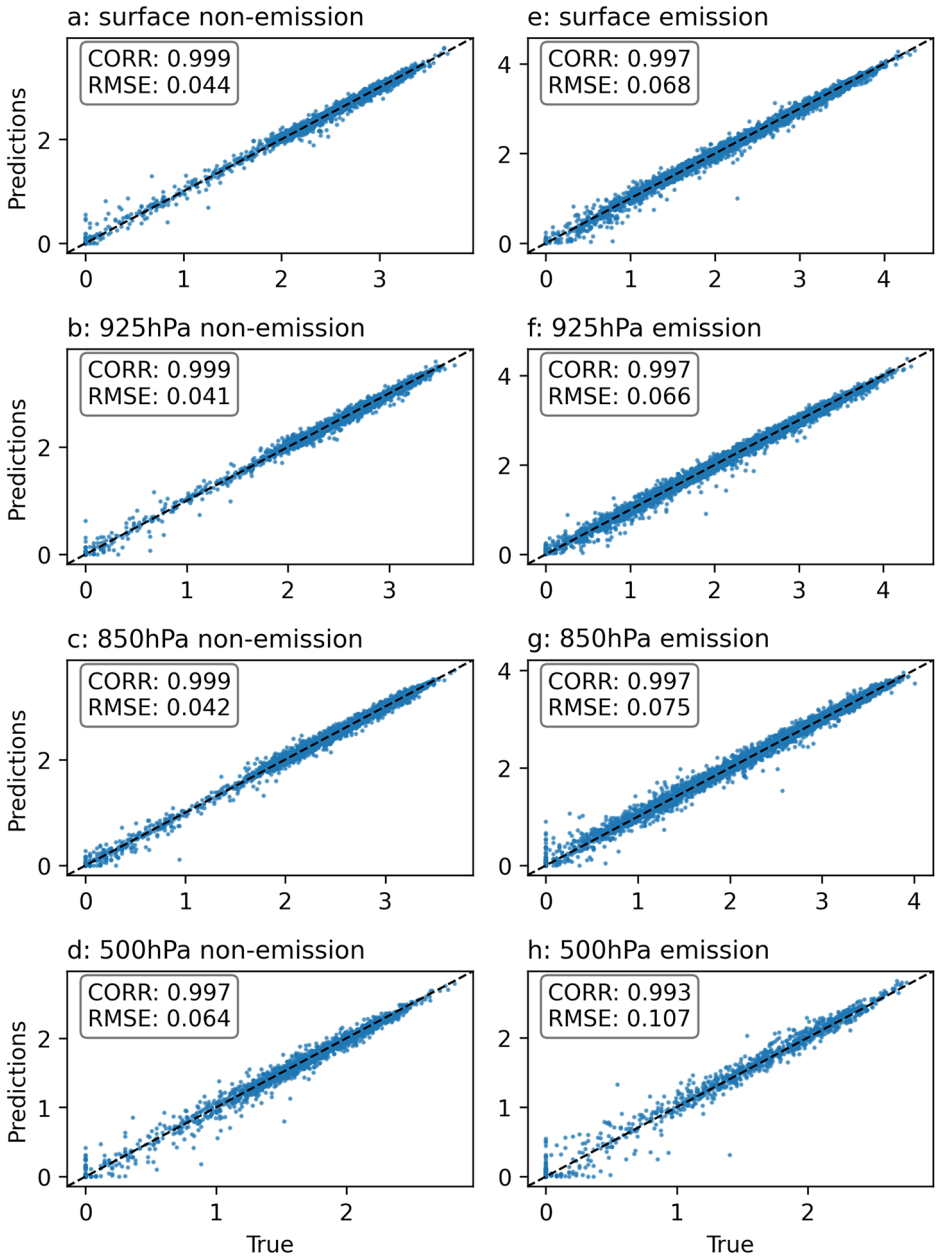


Fig. 1 | Comparison between observed and AI-DUST-predicted dust concentrations at the next time step. Observations are derived from dust event samples in 2024 across multiple vertical levels: **a**–**d** surface, 950 hPa, 850 hPa, and 500 hPa, respectively, for grid cells without dust emissions; **e**–**h** same levels for grid cells with active dust emissions. All data are in log-transformed values.

valuable insights for developing fully integrated AI-driven SDS forecasting systems.

In this work, we propose AI-DUST, a machine learning model designed to replace traditional physics-based dust transport modules. The model learns to represent key physical processes governing SDS dynamics, including transport, diffusion, and deposition, using only a limited set of vertical meteorological variables and without explicit boundary layer parameters. By doing so, it can be directly driven by AI-generated weather predictions. This integration not only eliminates the need for computationally expensive NWP simulations but also benefits from the improved forecast skills of AI-based weather models. We evaluate its performance using spring 2025 SDS events in East Asia and compare it against traditional numerical modeling systems.

Results

Single-step and multi-step forecast evaluation using NWP

To evaluate the single-step simulation performance of the AI-DUST model, we initialized it with dust emissions and concentrations from a traditional numerical model at a given time step. The same meteorological fields were used to drive both models for predicting the concentration changes in the subsequent time step (Fig. 1).

The results show that the AI-DUST model can effectively replicate the dust diffusion, transport, and deposition processes simulated in the traditional model. Dust concentrations showed an R exceeding 0.99. Notably, performance was higher in non-emission regions compared to emission-active regions, likely due to the absence of uncertainties associated with the vertical lofting of dust into the free atmosphere. Predictions at lower atmospheric levels outperformed those at higher altitudes, as near-surface dust concentrations are typically higher and more dynamically variable, providing richer training signals for the AI model. For instance, in non-emission regions, the R for near-surface dust concentrations reached 0.9994, with an RMSE of 0.0444. In contrast, in emission-active regions at 500 hPa, the R was 0.9927, with an RMSE of 0.1068. Despite these differences, overall performance across various regions and altitudes remained robust and consistent, indicating that the model effectively captures the dominant processes in diverse conditions.

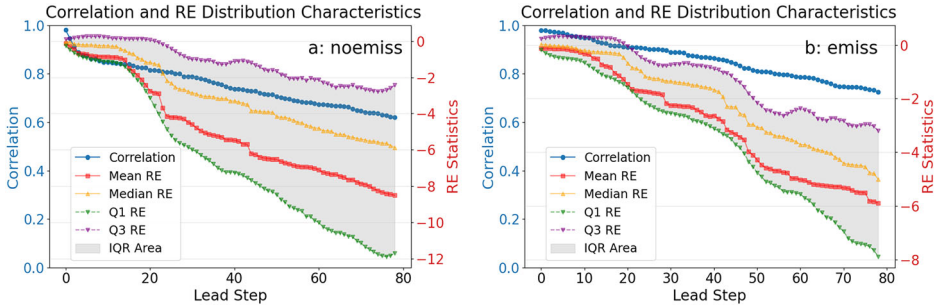


Fig. 2 | Temporal evolution of R (blue lines) and RE (gray area) between AI-DUST and the reference numerical model across forecast steps. **a** Surface dust concentrations in regions without dust emissions; **b** Surface dust concentrations in regions with active dust emissions.

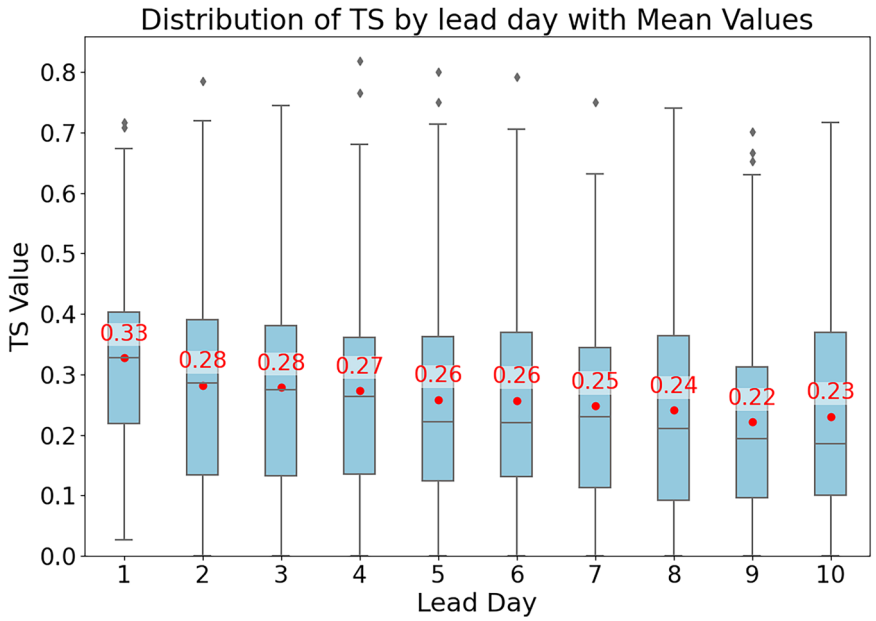


Fig. 3 | Daily TS of the AI-DUST model system for SDS weather forecasting over East Asia across 1–10 day forecast lead times. The red numbers indicate the average TS for each lead day.

To evaluate the long-term forecasting capability of the AI-DUST model, we conducted multi-step predictions up to 80-time steps ahead (Fig. 2). While performance gradually degraded with increasing forecast lead time, the model maintained physically consistent and skillful predictions throughout the simulation period. Even at the 80th step, the R remained above 0.6. In emission-active regions, the R reached 0.71 at the final step, with an RE of −5.9%. In non-emission regions, the R was 0.63, with a mean RE of −8.3%. The relatively weaker skill in non-emission regions is primarily attributed to the cumulative propagation of errors in dust transport, deposition, and diffusion processes, which are not reset by local emissions. Conversely, in emission-active regions, ongoing dust emissions act as a recurring source that partially offsets the accumulation of transport and deposition-related errors. Each step's emission contribution resets the error accumulation, leading to relatively stable long-term forecasts despite initial higher one-step errors.

Overall, the AI-DUST model demonstrates robust performance in both one-step and multi-step simulations. The results indicate strong model skill in reproducing dust concentration changes in a traditional numerical model under both emission-influenced and advection-dominated conditions.

Real-time forecast for SDS weather in East Asia 2025

Since 1 March 2025, we have been conducting operational SDS forecasts over East Asia using the AI-DUST model driven by AIFS from ECMWF, providing 10-day-ahead predictions. Model performance was evaluated against observational records during the main dust season (March–May 2025).

The model achieves a TS of 0.33 for 1-day ahead SDS event forecasts (Fig. 3). For days 2–3, the TS remains high at 0.28, gradually decreasing thereafter but staying above 0.22 throughout the entire 10-day forecast window. Notably, these TS outperform the typical performance of NWP operating in the WMO SDS-WAS Asian regional center. Most traditional models achieve TS values between 0.20 and 0.29 for 1–3 day forecasts[22], indicating that the 10th-day forecast from AI-DUST performs comparably to, or even surpasses, the 3rd-day forecast of some traditional numerical models.

To further evaluate the performance of the AI-DUST model, we conducted an independent validation using ground-based PM_{10} observations from the national air quality monitoring network. Figure 4 presents the spatial distribution of the daily R and RMSE between predicted and observed PM_{10} concentrations over the 1–10 day forecast lead times. On day 1, the model achieves a domain-averaged spatial R of 0.57, with an RMSE of 289 μg/m³. The R remains above 0.35 for lead times up to day 5, and even at day 10, it stays above 0.30, indicating sustained skill in capturing the spatiotemporal evolution of dust pollution. The RMSE gradually increases with forecast horizon but remains below 390 μg/m³ on average across the domain. These results demonstrate improved predictive accuracy compared to traditional numerical dust models[11].

During spring 2025, a total of 14 widespread SDS events occurred, all of which were used to evaluate the forecasting performance of the AI-DUST model. To benchmark its forecast skill in providing early warnings, we compared the TS of AI-DUST against those from the KMA model, a widely recognized numerical model known for its stability and accuracy in East Asia.

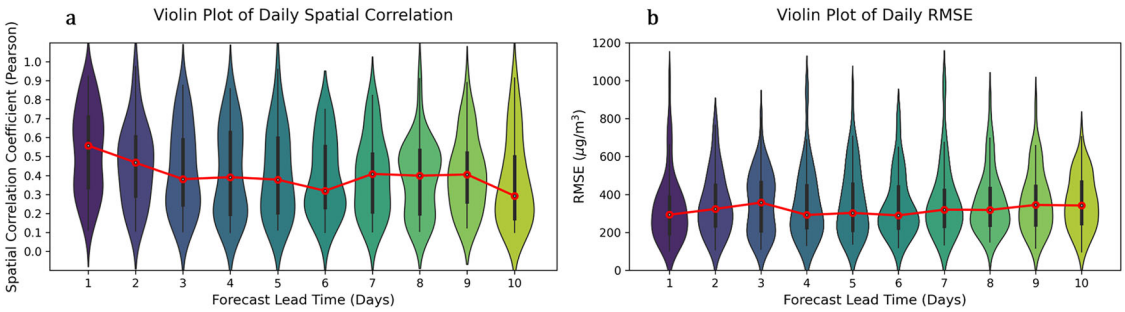
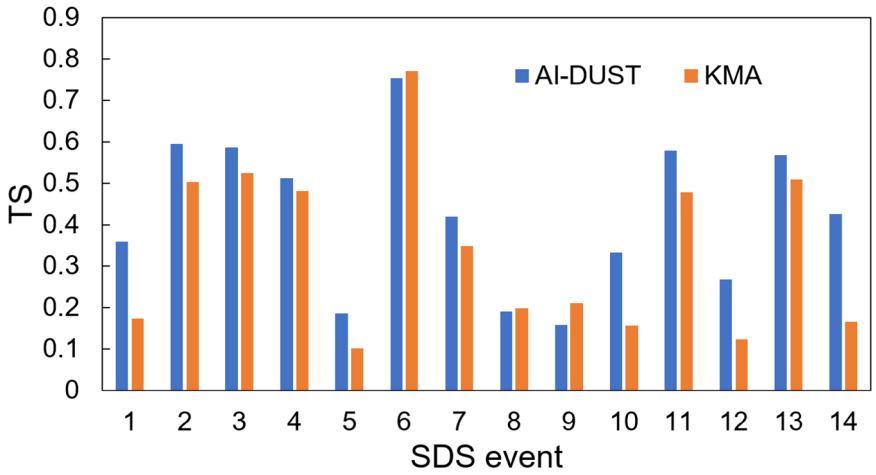


Fig. 4 | Spatial R and RMSE between AI-DUST predicted and observed PM_{10} concentrations over East Asia. a Spatial R and **b** RMSE. Metrics are averaged over the study region and shown for each forecast lead time from day 1 to day 10.

Fig. 5 | Comparison of TS between the AI-DUST and KMA models for 14 SDS events during spring 2025. The forecast data are evaluated at the 24–48-h lead time.



The results show that the AI-DUST model outperformed the KMA model in the majority of these events. The AI-DUST achieved an average TS of 0.42 over the 24–48 h forecast window, representing a 27% improvement relative to KMA (0.33). This significant enhancement demonstrates the competitive advantage of the AI-DUST model over traditional numerical models in regional SDS events forecasting.

Furthermore, as shown in Fig. 5, the TS for 9 out of the 14 events exceeded the seasonal mean (0.28) (Fig. 3), indicating consistently strong performance across strong SDS events. The model exhibited particularly high skill in strong SDS events capturing, highlighting its reliability for operational early warning systems.

Forecast for strong and severe SDS events in East Asia 2025

We conducted a more detailed analysis of the forecasting performance for four significant SDS events that reached central and eastern China (Fig. 6). These events were selected due to their large spatial extent and high intensity. In the source regions of northwest China, the model showed strong temporal correlation with observations, with R between predicted and observed PM_{10} concentrations exceeding 0.6 at most stations. RE were mostly within ±50%, indicating that the model reasonably captures dust emission dynamics in these areas. In downstream regions, central and eastern China, where dust is transported from the source areas, temporal correlations remained above 0.4, with RE ranging from 0 to 50%, suggesting a moderate underestimation of PM_{10} concentrations during long-range transport. This is consistent with the analysis presented in Section 3.1.

Comparisons with satellite observations (Fig. 7) further demonstrate that AI-DUST accurately reproduces the spatial extent and location of the SDS events. At the 48-h forecast lead time, the model achieved a spatial R of 0.63 for these four events, surpassing both the seasonal mean spatial correlation for 2-day forecasts (0.47) and even the seasonal mean for 1-day forecasts (0.57; Fig. 4a). This highlights the model's enhanced capability in simulating the spatial evolution of severe SDS events, particularly under extreme conditions.

Notably, the SDS event from April 10 to 14 was the first in nearly a decade to reach southern China, making it one of the most unusual dust events in recent decades. Despite the absence of similar events in the model's training data, AI-DUST successfully predicted both the spatial extent and intensity of this extreme episode. For this event, the model achieved a domain-averaged TS of 0.75. Over half of the observational stations in the affected region exhibited temporal R above 0.8 between predicted and observed PM_{10} concentrations, with RE confined to ±50% (Fig. 6c, g). In southern China specifically, PM_{10} concentration RE remained within ±20%, indicating high accuracy and low systematic error in this atypical region.

This successful prediction underscores the model's strong generalization capability, particularly for rare and extreme dust storms. It highlights the potential of AI-DUST for operational forecasting of unprecedented SDS events.

Model sensitive to meteorology fields and dust emissions

We designed a set of sensitivity experiments (Supplementary Note 2) to evaluate the performance of AI-DUST in simulating dust events during spring 2025. The model was driven by different meteorological forecast fields and surface characteristics over dust source regions to demonstrate its ability to represent key atmospheric physical processes of dust particles and to understand the reasons behind the improved forecast skill observed in our system. In addition to the control run driven by AIFS, we performed parallel simulations using meteorological inputs from the National Centers for Environmental Prediction Global Forecast System (NCEP-GFS) (downscaled by WRF) and the CMA global forecast system (CMA-GFS) to assess the impact of driving meteorology on SDS forecasts.

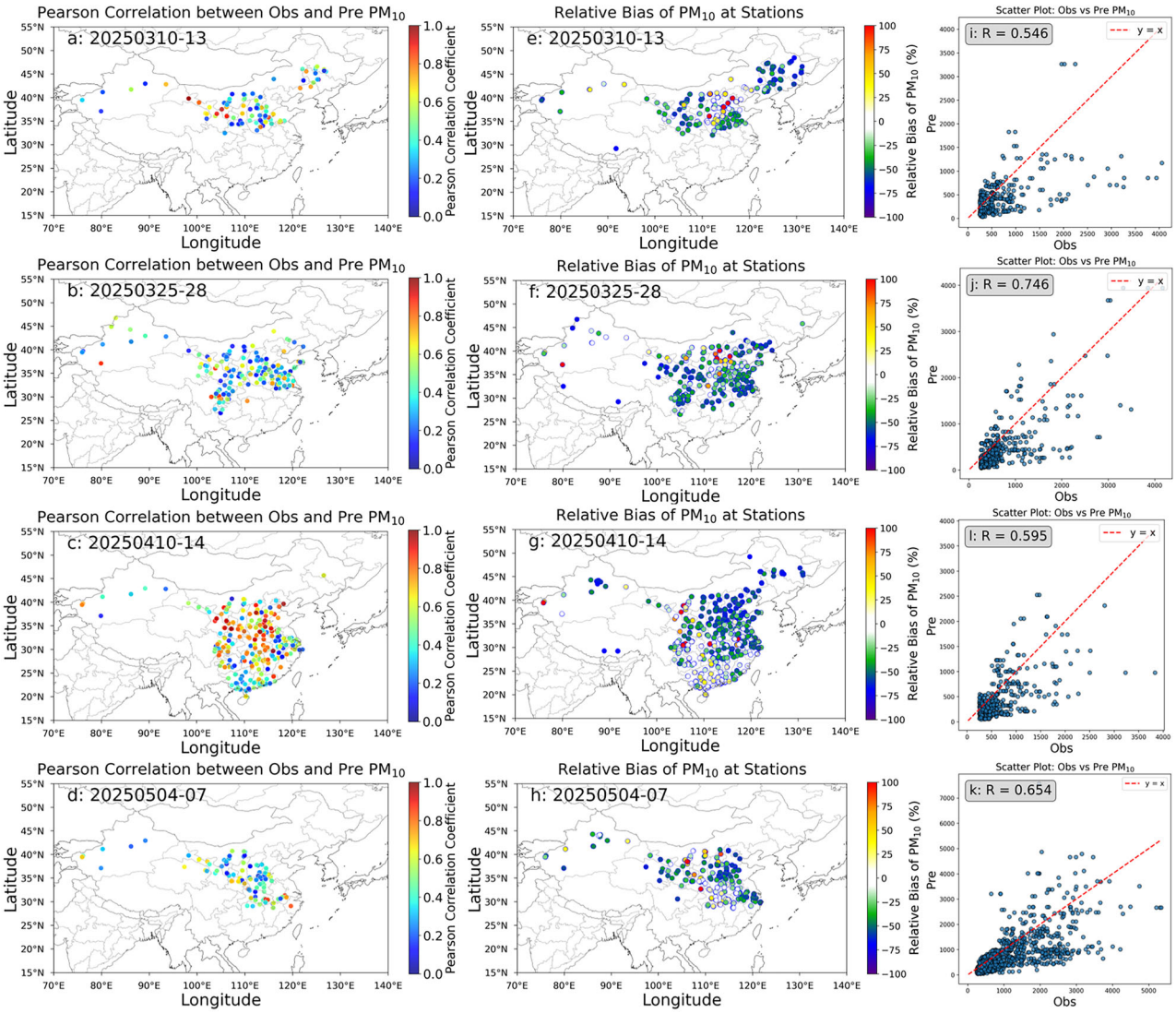


Fig. 6 | Evaluation of the AI-DUST model's performance for four major dust events during spring 2025. a–d show the temporal R of PM_{10} concentrations at different stations. **e–h** present the RE in peak PM_{10} concentration forecasts. **i–l** display the spatial R for each event.

As shown in Fig. 8, the TS of the three experiments exhibits clear differences. On day 1, the average TS is 0.33, with the lowest value at 0.22. By day 10, the average TS range from 0.23 to 0.18. To quantify the divergence among the simulations, we computed the RMSE of dust concentrations between the NCEP-driven and CMA-driven runs relative to the AIFS-driven reference. The RMSE exceeds 100 μg/m³ and increases with forecast duration, consistent with the behavior observed in traditional multi-model dust forecasting systems[11]. These results confirm that our model effectively captures the influence of different meteorological fields drivers on dust prediction (Fig. 9).

We also conducted experiments using different surface roughness densities in dust source regions. Results show that employing high-resolution surface roughness densities (Fig. 10) significantly improves the model's forecasting skill, with an average 18% increase in TS over forecast days 1–10. The enhancement is consistent across all lead times, indicating that refined surface data consistently benefits prediction accuracy. This also demonstrates that the AI-DUST model is sensitive to dust emission processes.

The sensitivity experiments demonstrate that the AI-DUST model effectively simulates the atmospheric physical processes of dust particles and exhibits clear sensitivity to both meteorological forcing fields and dust emissions. Among all experimental configurations, the highest TS are consistently achieved when the model is driven by AIFS meteorological forecasts and high-resolution surface characteristics. The superior performance of AI-DUST can thus be attributed to three key factors: its ability to accurately represent dust-related atmospheric physics, the high quality of AIFS meteorological forecasts, and the use of high-resolution surface roughness density that better constrains dust emission processes.

SDS Forecasts over North Africa and the Arabian Peninsula

To evaluate the geographical generalization capability of the AI-DUST model beyond its training domain in East Asia, we conducted zero-shot forecasting experiments for SDS events in North Africa and the Arabian Peninsula. Figure 11 compares AOT from VIIRS Deep Blue with AI-DUST predictions for 22 March and 12 May 2025.

On 22 March, a prominent dust plume extended from the central Sahara (0°–20°E) northward, as evidenced by high AOT in satellite observations. Despite data gaps due to partial cloud cover, the spatial structure of this northward-traveling SDS band was clearly identifiable. Additionally, high AOT was observed near 30°E in the southern Sahara. AI-DUST successfully reproduced the SDS feature with strong spatial agreement. However, it slightly overestimated SDS in the 30°E region, indicating possible limitations in local emission parameterizations.

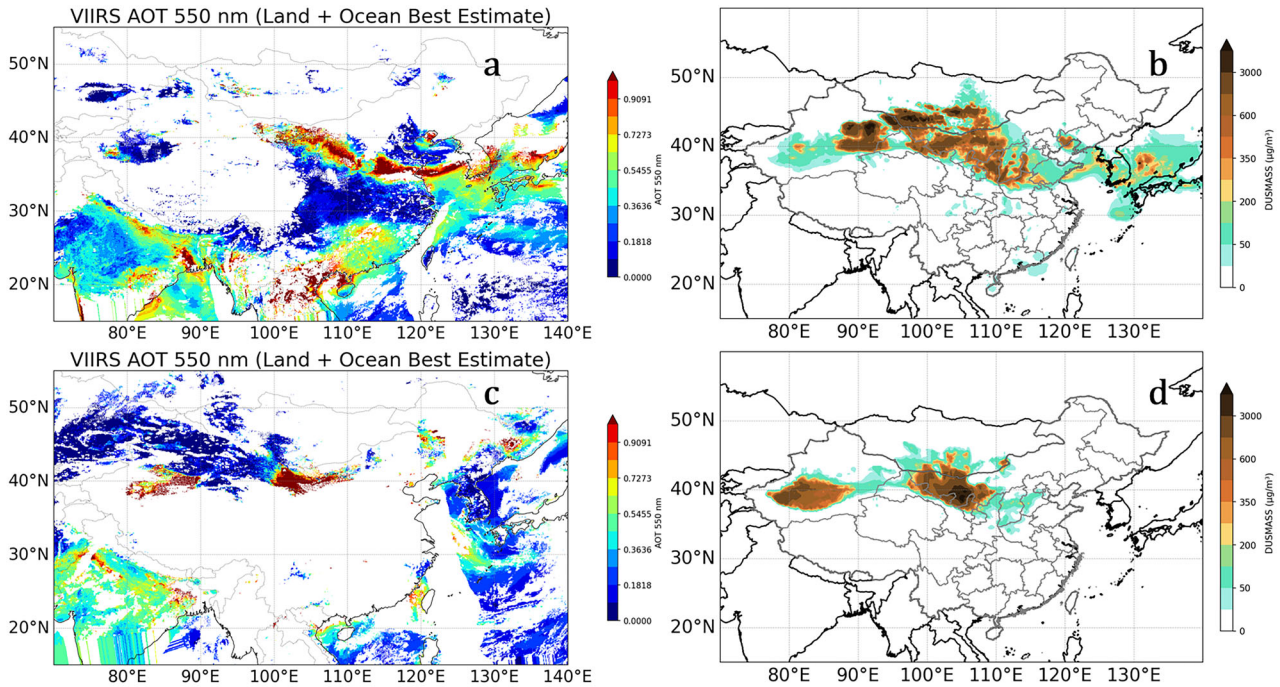
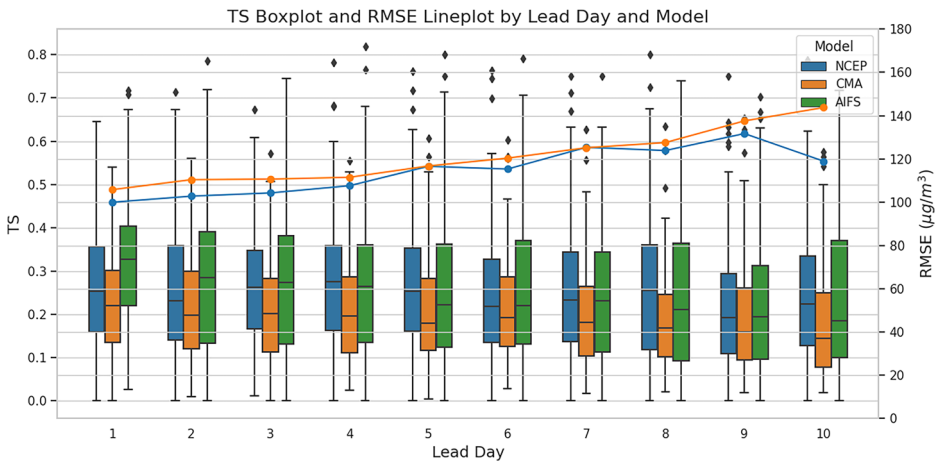


Fig. 7 | Comparison between satellite-observed AOT and AI-DUST predicted dust concentration. show the AOT data, while **b**, **d** display the AI-DUST predictions at coincident times on 25 March (**a**, **b**) and 11 April (**c**, **d**) 2025. Satellite observations were acquired at local times of 11:00, 12:00, 14:00, 15:00, and 16:00. The AI-DUST model forecasts are sampled at the nearest forecast time step for each grid cell to match the satellite overpass time.

Fig. 8 | Evaluation of AI-DUST dust forecasts driven by different meteorological fields. The boxplots show the TS across forecast lead times for AI-DUST simulations driven by NCEP (blue), CMA (orange), and AIFS (green). The line plots display the RMSE of dust concentrations from the NCEP-driven (blue) and CMA-driven (orange) simulations relative to the AIFS-driven reference.



On 12 May, intense SDS activity occurred across two major zones, the northern and western Sahara, and the northeastern Arabian Peninsula. The AI-DUST model accurately reproduced the location and spatial extent of both SDSs, demonstrating strong predictive skill despite the geographic and meteorological differences from its training domain.

Overall, the model's performance in these subtropical and tropical arid regions is highly encouraging. Its ability to reproduce SDS features without retraining or regional tuning demonstrates robust generalization and geographical portability. This cross-regional success suggests that AI-DUST has learned fundamental physical patterns governing dust diffusion, transport, and deposition, rather than merely memorizing region-specific statistics, highlighting its potential as a globally applicable tool for SDS forecasting.

Discussion

This study presents an AI-driven dust forecasting model, AI-DUST, directly forced by AI-based weather forecast products. The model effectively replaces key physical processes, such as transport, diffusion, and deposition, traditionally simulated by numerical atmospheric models. Previous AI-based atmospheric chemistry models often rely on mesoscale meteorological models to provide meteorological fields, which increases computational cost and limits the full utilization of the AI-generated weather forecasting skill of AI weather models. To bridge this gap, we propose a deep learning framework that integrates multiple stacked Graph Attention Network with physical constraints, further enhanced by a physically informed dust emission scheme to simulate and predict dust events.

Evaluation of the AI-DUST model demonstrates robust performance in reproducing the simulations of traditional dust models. Compared with the traditional model, the one-step-ahead prediction achieves a temporal R exceeding 0.99, and the R remains above 0.6 even at an 80-step lead time, with RE within 0-10%.

In real-time forecasts of spring 2025 dust events over East Asia, AI-DUST outperforms the operational models used by the WMO SDS-WAS

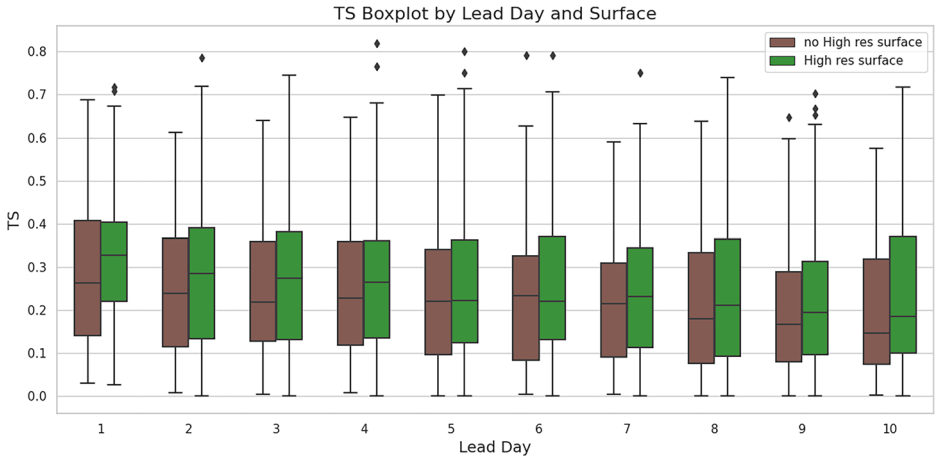


Fig. 9 | TS using different surface roughness density datasets. Simulations without high-resolution surface roughness density are shown in brown, while those utilizing enhanced data are depicted in green.

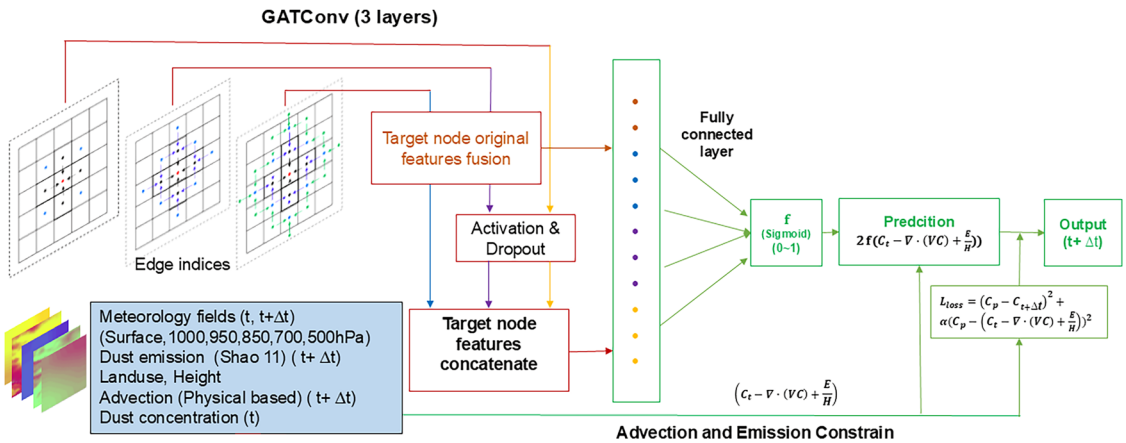


Fig. 10 | Architecture of the deep graph network for next time dust concentration forecast. The model operates on a 15 × 15 grid graph with four-connected nodes. Stacked GAT layers aggregate neighborhood information, with feature fusion from advection, emission, and initial concentration introduced after the first layer. Final predictions are generated via feature concatenation, a fully connected layer, and Sigmoid activation. Physical consistency is enforced through prediction limitation and mass conservation-aware loss. Separate models are trained for each layer, treating the concentration at one specific layer as the prediction target at a time. Training is further stratified by surface emission activity (emission vs. non-emission zones).

(Beijing) center. For 48-h forecasts across 14 SDS events, AI-DUST increases the TS by 27% relative to the traditional model. Notably, its 10-day forecasts achieve a TS above 0.22, surpassing the typical 3-day forecast skill of conventional numerical models. This unprecedented skill in long-term dust prediction underscores the model's ability to capture complex atmospheric transport and emission processes.

The high predictive accuracy of the AI-DUST model can be attributed to three synergistic factors. First, the model explicitly simulates the physical processes of dust emission, transport, and deposition through a physics-informed deep learning architecture. Second, AI-DUST leverages the forecasting capability of the AI-based weather forecasts to accurately capture the driving meteorological fields, without relying on external NWP models. Third, the dust emission scheme incorporates high-resolution, regionally representative surface characteristics, including surface roughness that governs dust erodibility in East Asia.

Beyond accurately predicting extreme SDS events in the training domain, AI-DUST successfully simulates SDSs in untrained regions, including northern Africa and the Arabian Peninsula, demonstrating strong generalization and portability. This capability is largely attributed to the standardized preprocessing of training data and special model architecture, which enables the model to extract location-invariant, physically meaningful features and perform reliably under diverse environmental conditions.

The development of AI-DUST provides a promising paradigm for leveraging AI weather forecast products in downstream environmental modeling. Future work will focus on integrating additional AI-based modules for atmospheric chemistry and aerosols into this framework, aiming to construct a comprehensive global atmospheric chemistry model driven directly by AI weather forecasts. This approach avoids redundant training on meteorological fields and instead capitalizes on the latest advances in global AI-generated weather forecasts, allowing researchers to concentrate on improving the representation of chemical and aerosol processes.

Methods

AI based SDS forecast model

A deep graph neural network (AI-DUST) is proposed to replace the transport, diffusion, and dry deposition modules in traditional dust numerical modeling, enabling seamless integration with AI-generated weather forecasts for 3-h sand and dust predictions. The model combines graph convolution, feature fusion, and explicit feature constraints within a framework designed for graph-structured atmospheric data (Fig. 10). It incorporates multiple layers of Graph Attention Networks (GATs) with feature fusion mechanisms at each layer, explicitly regularizing the target node features to ensure physical consistency. The network operates on a 15 × 15 grid graph, where each node is connected to its four spatial neighbors, and the central node serves as the prediction target.

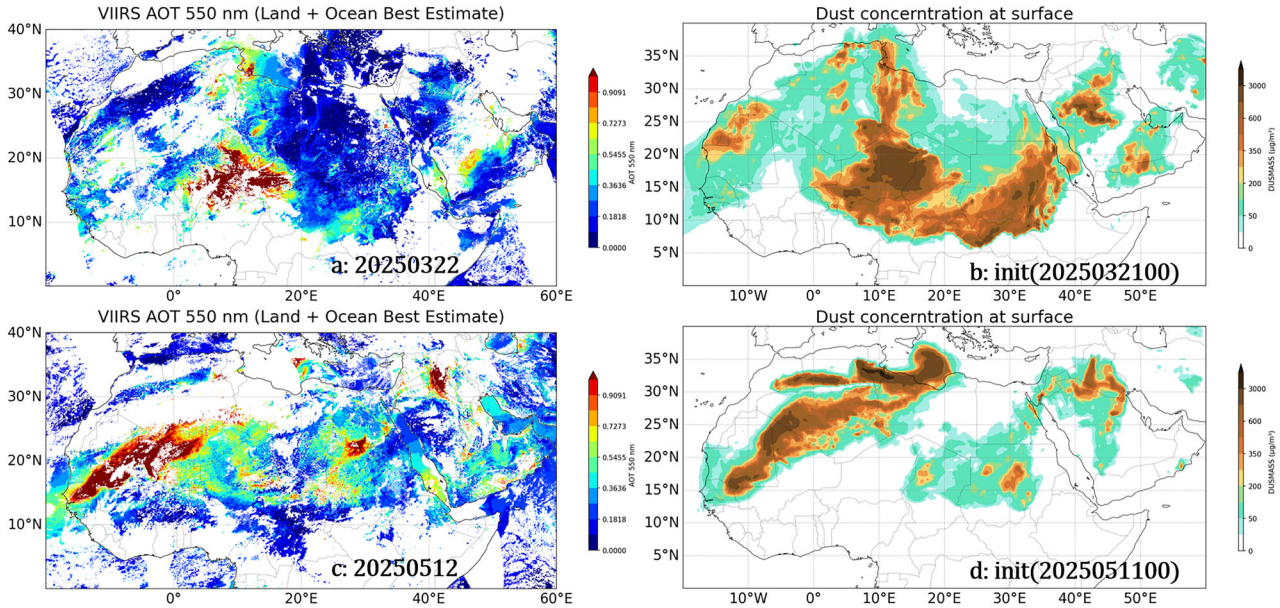


Fig. 11 | Comparison of satellite-observed AOT and AI-DUST predicted dust concentration over North Africa and the Arabian Peninsula on 22 March and 12 May 2025. a, **c** show VIIRS Deep Blue AOT observations; **b**, **d** show AI-DUST forecasts initialized at 00:00 UTC on 21 March and 11 May, respectively. Model forecasts are sampled at times closest to satellite overpass for direct comparison.

Multiple stacked GAT[23]. layers aggregate neighborhood information through learnable attention weights and have been successfully applied to forecasting atmospheric pollutants[24,25]. The model adopts a staged feature integration strategy: the first GAT layer processes the raw node features using standard graph convolution without additional feature fusion. Starting from the second layer, a feature fusion mechanism is introduced to enhance the representation of key physical inputs, advection, dust emission, and initial concentration. At each subsequent layer, separate learnable linear projections are applied to the target node's features, and the enhanced outputs are concatenated with the graph convolution results to form enriched node embeddings. These fused representations serve as input to the next layer. All GAT layers operate on the evolving node embeddings, with hidden layers employing LeakyReLU activations and dropout regularization to improve generalization.

To embed physical consistency directly into both the architecture and the learning objective, we use three complementary components. First, the model combines features from all convolutional layers with the original input key features at each location. This multi-scale representation is then passed through a final fully connected layer to produce a richer, more informative output. Second, instead of predicting dust concentration directly, the network outputs a value α between 0 and 1 using a sigmoid function. This value scales a physically based reference concentration (defined in Eq. 1) to produce the final prediction. This ensures predictions are non-negative, bounded, and consistent with expected dust behavior. Third, the model is trained with a loss function that includes both data accuracy and a physics-based constraint (defined in Eq. 2). Together, these elements allow the model to learn from data while staying grounded in the basic physics of dust transport.

$$C_p = 2f(C_t + (-\nabla \cdot (VC) + E/H)) \tag{1}$$

$$L_{loss} = \left(C_p - C_{t+\Delta t}\right)^2 + \alpha(C_p - (C_t - \nabla \cdot (VC) + \frac{E}{H}))^2 \tag{2}$$

$$\alpha = \begin{cases} 0, & 0.1 < C_p/(C_t - \nabla \cdot (VC) + \frac{E}{H}) < 10 \\ 10, & otherwise \end{cases} \tag{3}$$

C_p denotes the predicted air pollutant concentration at the next forecast time step, C_t the observed dust concentration at the current time, and $C_{t+\Delta t}$ the corresponding ground-truth concentration in the training data. The scale factor f is between 0 and 1, and is generated from the Sigmoid function. To ensure that dust concentrations can be positively adjusted, f is multiplied by 2 in Eq. 1. The advection term, denoted as $\nabla \cdot (VC)$[26]. represents the mass transport due to wind fields and spatial gradients of dust concentration from time t to $t + \Delta t$. The emission contribution is modeled as E/H, where E is the dust emission flux, and H is the assumed mixing layer height (set to 3000 m), following typical parameterization in dust modeling. The weighting coefficient α (defined in Eq. 3) is adaptive: it increases when the discrepancy between C_p and the sum of $C_t - \nabla \cdot (VC) + \frac{E}{H}$ is large. Here, $C_t - \nabla \cdot (VC) + \frac{E}{H}$ represents a simplified physics-based estimate of dust concentration. This approach helps keep the model's predictions within a realistic range based on physics, without making training unreasonable.

This architecture uniquely combines graph-based relational learning with explicit constraint enforcement, making it particularly suitable for structured prediction tasks where output values must maintain specific relationships with input features.

Training and testing dataset

To generate meteorological, geographic, and dust concentration features for training, validating, and testing the AI-DUST model, we conducted a 5-year simulation of dust events over East Asia (2020–2024) using the Weather Research and Forecasting (WRF) model[27] and the Comprehensive Air Quality Model with Extensions (CAMx). The set of model systems is provided in Supplementary Note 1.

The modeling domain covers major dust source and downwind regions in East Asia (Fig. 12), spanning 70°E-140°E and 10°N-55°N. Simulations were performed on a regular latitude-longitude grid with a horizontal resolution of 0.25°. The dust emission scheme used in our work based on the friction velocity and threshold friction velocity[28]. The threshold friction velocity (u_{*t}) for dry and smooth surfaces depends on soil particle size and air density. The soil texture datasets are derived from Hengl[29]. The grain size follows Wei et al.[30] and Gong et al.[31]. Normalized Difference

Vegetation Index (NDVI) dataset (MOD13Q1)[32] were used to estimate the vegetation cover.

We interpolated the simulations from the NWP to obtain meteorological fields and dust concentration at the surface and selected pressure levels (surface, 1000, 950, 850, 700, and 500 hPa), which were used to construct the training dataset (Table 1). The data were organized into 15 × 15 grid-point samples centered on a target node, resulting in a large sample pool exceeding 200 million instances. For grid points near the boundary of the domain, we pad the missing neighboring values with zeros.

To reduce the total number of training samples while preserving representative distributions across distinct physical regimes. We applied a stratified random sampling with the following criteria: (1) The number of samples with and without dust emission sources at the central grid cell is equal. (2) Samples with significant elevation differences, defined as absolute relative height > 500 m between the center and surrounding grids, accounted for at least one-fifth of the dataset. (3) We ensured an equal number of samples across predefined dC ($C_{t+\Delta t} - C_t$) intervals: for emission-active cases (<500, 500–1000, 1000–3000, >3000 μg/m³) and emission-inactive cases (<100, 100–500, 500–1000, >1000 μg/m³). Similar data subsampling approaches are widely adopted in large-scale machine learning tasks to improve training efficiency[33].

The final training dataset comprises simulations from 2020 to 2023, with separate models trained for dust concentration at each vertical level. The year 2024 is reserved for independent testing to evaluate model generalization performance.

To account for the spatial variation in grid cell size due to the latitude-longitude projection, the horizontal wind components are normalized by the grid spacing in their respective directions. The standardized zonal and meridional wind components are computed as Eqs. 4 and 5. To decouple terrain representation from absolute positioning, terrain elevation is represented as relative height with respect to the center grid point. By removing absolute elevation values, this representation helps avoid overfitting to terrain-specific altitude patterns[34]. Specifically, the elevation at the central node is set to 0, and all surrounding grid points are adjusted relative to this reference, effectively removing absolute location bias. All meteorological variables (including wind speed, temperature, etc.) and relative elevation data were normalized to zero mean and unit variance. Dust concentration and emission flux, both characterized by high dynamic range and skewed distributions, were transformed using the natural logarithm (ln) to stabilize variance and improve model learning efficiency. This preprocessing pipeline ensures that the model learns spatial patterns and physical relationships rather than memorizing absolute geographic or scale-dependent features.

$$u_s = u/dx \tag{4}$$

$$v_s = v/\mathrm{d}y \tag{5}$$

where u and v are the original zonal and meridional wind velocities (in m/s), and dx, dy are the zonal and meridional grid widths (in meters). Here, dx is evaluated at the center of each grid cell to reflect local resolution, while dy is approximately constant due to the uniform latitude spacing. This normalization reduces scale dependence across latitudes and ensures consistent representation of wind-driven transport in the model input.

To avoid interference from mixed emission dynamics and improve model specialization, we adopt a separate training strategy for samples with and without dust emissions. Specifically, two distinct models are trained: one dedicated to emission-active regions and another for emission-inactive regions. During inference, the appropriate model is selected based on whether the central grid cell is identified as an active emission source. This separation allows the model to better capture the distinct physical processes governing dust concentration changes in each scenario, such as emission-driven bursts versus advection-dominated transport, without mutual bias.

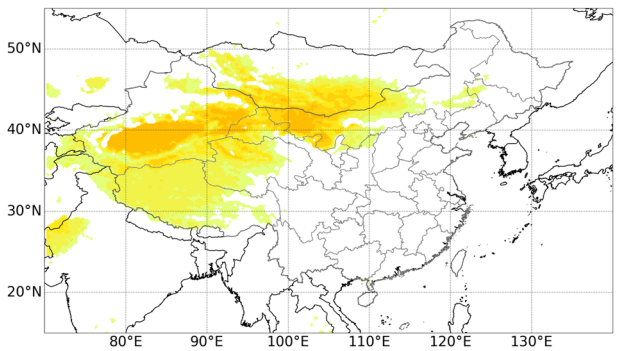


Fig. 12 | The model domain and the distribution of dust sources. The yellow region is the dust source region.

Real-time forecast and validation

We employ the products form Artificial Intelligence Forecast System (AIFS) from the European Centre for Medium-range Weather Forecasts (ECMWF) to drive the AI-DUST model for real-time SDS forecasting during spring 2025. AIFS is a AI-generated weather forecasts with a horizontal resolution of 0.25°, providing forecasts of meteorological variables at the surface and multiple vertical levels, as well as soil moisture, a key state variable for the dust emission scheme[35]. These forecast fields are directly used to drive the AI-DUST model and the embedded dust emission scheme. This seamless integration enables the AI-DUST model for dust concentration forecasting without the need for additional downscaling or preprocessing.

Meteorological observations of SDS weather phenomenon and PM_{10} concentration data from spring 2025 were used to evaluate the forecasting performance of the AI-DUST model. The meteorological data were obtained from the China Meteorological Administration (CMA), and the surface PM_{10} concentration data were provided by the China National Environmental Monitoring Centre (CNEMC) (https://air.cnemc.cn:18007/). Aerosol optical thickness (AOT) from Visible Infrared Imaging Radiometer Suite (VIIRS) board the Suomi National Polar-orbiting Partnership (SNPP)[36] were also employed in this study. These AOT data provide critical validation for the spatial distribution and magnitude of dust aerosols predicted by the model.

Two traditional dust forecast models from the World Meteorological Organization SDS Warning Advisory and Assessment System (WMO SDS-WAS) Asian regional center were used for comparison with the AI-DUST model: the CMA and the KMA model. The CMA Unified Atmospheric Chemistry Environment model (CUACE) is a regional air quality forecasting system developed by Chinese Academy of Meteorological Sciences[37]. The Asian Dust Aerosol Model 3 (ADAM3), developed by the Korea Meteorological Administration (KMA) forecast Asian dust events[38]. Among

Table 1 | The specific meteorological and environmental variables used in the dataset

Input variables	Dust concentration	Wind	Temperature	Dust advections	Vegetable cover	Relative Height, Landuse, Surface roughness density
Time	t	t, t + Δt			t	Static
Layers	Surface, 950, 850, 500 hPa	Surface, 1000,950, 850,700 and 500 hPa			Surface	
Target variables	Dust concentrations at the surface and at 950, 850, and 500 hPa at time t + Δt					

the models evaluated in the WMO SDS-WAS Beijing Node, ADAM3 has consistently demonstrated superior performance[11].

During spring 2025, a total of 14 dust weather events were recorded in China. Notably, an exceptional dust event occurred from 10 to 14 April 2025, during which dust reached Southern China. Such a pronounced southward extension of dust transport is rare in the observational record.

Model performance for PM_{10} concentration predictions was evaluated using three standard metrics: relative errors (RE), root mean square error (RMSE), and Pearson correlation(R). To assess the accuracy of simulated SDS occurrences, the threat score (TS), defined in Eq. (5), was computed to quantify the spatial and temporal agreement between predicted and observed SDS weather phenomenon form meteorological observations.

$$TS = NC/(NF + NO + NC) \quad (5)$$

where, TS is the threat score, NC is the number of stations where the forecast for dust weather is correct, NF is the number of stations where the model forecast PM_{10} concentrations are greater than 200 μg/m[3,39], NO is the number of stations where the dust weather occurs.

Data availability

The real-time forecast generated by AI-DUST are publicly accessible through the WMO SDS-WAS Asian regional center platform at: http://www.asdf-bj.net/publish/forecasts/fq/ea/surface-concentration.html.

Code availability

The source code for AI-DUST model, along with trained model parameters, is available under an open-source license at https://gitee.com/hitwjk/ai-dust/.

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Author contributions

J.W. processed data, designed the models, and wrote the original draft. H.C. supervised the research. All coauthors helped improve the paper.

Competing interests

The authors declare no competing interests.

Additional information

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